EE4-27
EE9-CS5-2
EE9-FPN2-08

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

MSc and EEE/EIE PART IV: MEng and ACGI

SYSTEMS IDENTIFICATION

Corrected copy

Thursday, 17 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Given a stationary stochastic process $v(\cdot)$ with $\mathbb{E}(v) = 0$, consider its spectrum $\Gamma_v(\omega)$ which takes on the following values:

$$\Gamma_v(0) = \frac{9}{4}, \quad \Gamma_v(\pi/2) = \frac{5}{4}, \quad \Gamma_v(\pi) = \frac{1}{4}, \tag{1.1}$$

a) Consider the moving-average stationary stochastic process $w(\cdot)$ given by

$$MA(1): \; w(t) = e(t) + ce(t-1)$$

where $c \in \Re$, $c \neq 0$, $|c| < 1$, and $e(\cdot) \sim WN(0, \sigma_e^2)$.

Compute all the pairs of parameters (c, σ_e^2) such that

$$\Gamma_w(0) = \Gamma_v(0), \quad \Gamma_w(\pi/2) = \Gamma_v(\pi/2), \quad \Gamma_w(\pi) = \Gamma_v(\pi).$$

where $\Gamma_w(\omega)$ denotes the spectrum of the stochastic process $w(\cdot)$ and where $\Gamma_v(0), \Gamma_v(\pi/2), \Gamma_v(\pi)$ take on the values given in (1.1).

[7 Marks]

b) Determine the analytical expression of the correlation function $\gamma_w(\tau), \forall \tau$ of the stationary stochastic process $w(\cdot)$ defined in Question 1-a) using the pairs of parameters (c, σ_e^2) determined in your answer to Question 1-a).

[5 Marks]

c) Consider the auto-regressive stationary stochastic process $y(\cdot)$ given by

$$AR(1): \; y(t) = ay(t-1) + \eta(t)$$

where $a \in \Re$, $a \neq 0$, $|a| < 1$, and $\eta(\cdot) \sim WN(0, \sigma_\eta^2)$.

Compute, if possible, values of the parameters a and σ_η^2 such that

$$\Gamma_y(0) = \Gamma_v(0), \quad \Gamma_y(\pi/2) = \Gamma_v(\pi/2), \quad \Gamma_y(\pi) = \Gamma_v(\pi).$$

where $\Gamma_y(\omega)$ denotes the spectrum of the stochastic process $y(\cdot)$ and where $\Gamma_v(0), \Gamma_v(\pi/2), \Gamma_v(\pi)$ take on the values given in (1.1). Discuss your findings.

[8 Marks]

2. Consider a stationary stochastic process $v(\cdot)$ with $\mathbb{E}(v) = 0$ and characterized by the complex spectrum

$$\Phi_v(z) = \frac{(z+1/4)(z+4)}{(z-2)(1/2-z)}$$

a) Determine a transfer function $H(z)$ such that the *stationary* stochastic process $v(\cdot)$ can be modelled as

$$v(t) = H(z)\xi(t)$$

where $\xi(\cdot)$ is a white stochastic process $\xi(\cdot) \sim WN(0,1)$.

[5 Marks]

b) Determine the difference equation yielding the optimal one-step ahead prediction $\hat{v}(t+1|t)$ of $v(t+1)$ on the basis of past values $v(t), v(t-1), v(t-2), \ldots$ of the process $v(\cdot)$.

[3 Marks]

c) Denoting by $\varepsilon_1(t) = v(t+1) - \hat{v}(t+1|t)$ the one-step ahead prediction error when using the predictor determined in your answer to Question 2-b), determine its model as a stochastic process; moreover compute $\mathbb{E}(\varepsilon_1)$ and $\text{var}[\varepsilon_1]$.

[3 Marks]

d) Determine the difference equation yielding the optimal three-step ahead prediction $\hat{v}(t+3|t)$ of $v(t+3)$ on the basis of past values $v(t), v(t-1), v(t-2), \ldots$ of the process $v(\cdot)$.

[3 Marks]

e) Denoting by $\varepsilon_3(t) = v(t+3) - \hat{v}(t+3|t)$ the three-step ahead prediction error when using the predictor determined in your answer to Question 2-d), determine its model as a stochastic process; moreover compute $\mathbb{E}(\varepsilon_3)$ and $\text{var}[\varepsilon_3]$.

[3 Marks]

f) Compare $\text{var}[\varepsilon_1(t)]$ computed in your answer to Question 2-c) with $\text{var}[\varepsilon_3(t)]$ computed in your answer to Question 2-e). Comment on your findings.

[3 Marks]

3. Consider the electrical circuit depicted in Fig. 3.1.

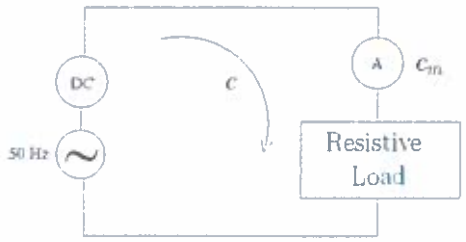


Figure 3.1 Electrical circuit with a resistive load and a DC combined with 50Hz AC power supply.

A current meter is available providing a measurement c_m of the current c flowing in the circuit.

An experiment to identify the DC bias and the phase of the current c is set up. A set of M given values of time-samples $t(j)$, $j = 1,\ldots,M$ are selected. Correspondingly, a set of M measurements of the current are recorded by the current meter:

$$c_m(j) = c(j) + \xi(j),\ j = 1,\ldots,M$$

where $c(j) = \gamma + K\sin[100\pi t(j) + \varphi]$ and ξ is a zero-mean white process $\xi(\cdot) \sim WN(0,\lambda^2)$.

a) Assuming the amplitude K to be known, devise a *parametric* model $\mathcal{M}$ of the unknown current $c(t)$ of the form

$$\mathcal{M}: \widehat{c}(t,\alpha) = \sum_{i=1}^{N} \alpha_i \rho_i(t)$$

for a suitable choice of N and of the functions $\rho_i(t)$, $i = 1,\ldots,N$, where $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_N]^\top$ is a vector of parameters to be determined using the available measurements. [*Hint: exploit the known form of the current $c(t)$ and the trigonometric properties on the* $\sin(\cdot)$ *function*].

[7 Marks]

b) Consider the parametric model $\mathcal{M}$ designed in your answer to Question 3-a). The parameters α_i, $i = 1,\ldots,N$ have to be determined using a set of time-instants and current measurements

$$\Theta = \{(t(j), c_m(j)),\ j = 1,\ldots,M\}.$$

Devise a least-squares method the solution of which provides the optimal (in the least-squares sense) sets of parameters $\alpha_i^\star$, $i = 1,\ldots,N$ and determine its general solution in analytical form.

[8 Marks]

c) Consider the following set Θ of 5 time-instants and corresponding measurements of current, where $K = 1$:

$$\Theta = \{(0, 0.65), (0.005, 1.16), (0.01, -0.08), (0.015, -0.33), (0.02, 1.07)\}$$

Using the general solution obtained in your answer to Question 3-b), compute the optimal parameters $\alpha_i^\star$, $i = 1,\ldots,N$. Sketch the plot of the time-behaviour of the obtained approximate model and also plot on the same diagram the 5 points of the set Θ. Discuss your findings.

[5 Marks]

4. Consider two different stochastic processes $v_1(\cdot)$ and $v_2(\cdot)$ generated as illustrated in Fig. 4.1.

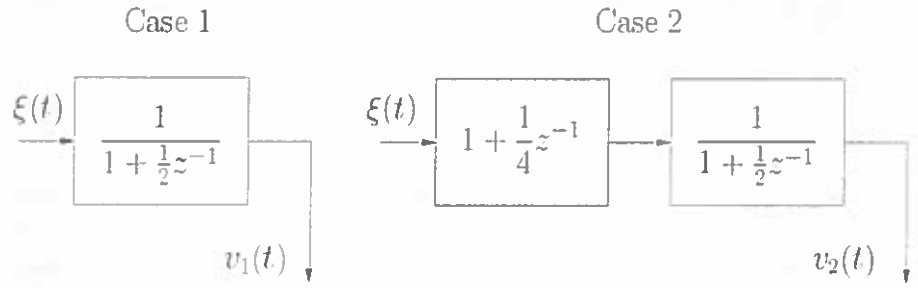


Figure 4.1 Two stochastic data-generation mechanisms.

Moreover, consider the family of AR(1) stochastic models

$$\mathcal{M}(\theta): \quad y(t) = ay(t-1) + \eta(t), \quad \theta := a$$

where, with reference to Case 1 in Fig. 4.1, $y(t) = v_1(t)$, whereas with reference to Case 2 in Fig. 4.1, $y(t) = v_2(t)$.

In both Case 1 and Case 2, consider an identification procedure in which $\widehat{a}(N)$ is the least squares estimate of a based on N measurements $\{y(1), y(2), \ldots, y(N)\}$.

a) **Case 1.** The stochastic process is $y(\cdot) = v_1(\cdot)$ with $\xi(\cdot) \sim WN(0,1)$. Determine the value $\bar{a}_1$ the estimate $\widehat{a}(N)$ approaches for large values of N (that is, $\bar{a}_1 = \lim_{N\to\infty} \widehat{a}(N)$, a.s.).

[6 Marks]

b) **Case 2.** The stochastic process is $y(\cdot) = v_2(\cdot)$ with $\xi(\cdot) \sim WN(0,1)$. Determine the value $\bar{a}_2$ the estimate $\widehat{a}(N)$ approaches for large values of N (that is, $\bar{a}_2 = \lim_{N\to\infty} \widehat{a}(N)$, a.s.).

[7 Marks]

c) With reference to Case 1 of Question 4-a) and Case 2 of Question 4-b), compute and compare the variances of the prediction errors

$$\text{var}[v_1(t) - \bar{a}_1 v_1(t-1)] \quad \text{and} \quad \text{var}[v_2(t) - \bar{a}_2 v_2(t-1)]$$

Comment on your findings.

Moreover, consider a different variance for the noise ξ, namely $\xi(\cdot) \sim WN(0.3)$. State whether or not the values of $\bar{a}_1$ and $\bar{a}_2$ computed in the answers to Questions 4-a) and 4-b) are different because of the different value taken on by the variance of ξ. Justify your answer.

[7 Marks]